Lab 09 Practice Problems

Question 1 [8 points]

Researchers are evaluating the effectiveness of a GLP-1 receptor agonist for weight management. In a randomized controlled trial, adults with obesity were assigned to receive either the GLP-1 receptor agonist or a placebo for 16 weeks. The primary outcome was whether each participant achieved at least 5% weight loss during the study period. The goal is to test whether the proportion achieving $\geq 5\%$ weight loss differs between the two groups.

To avoid discrepancies, DO NOT ROUND INTERMEDIATE VALUES (e.g., $\hat{p}$ or SE). Round only final answers as instructed.

Group	$\geq 5\%$ weight loss	$< 5\%$ weight loss	Total	$\hat{p}$
GLP-1 agonist	92	408	500	0.184
Placebo	31	469	500	0.062

Choose the appropriate test. You may assume all assumptions are met.

1.1 [2 points] State the null and alternative hypotheses for this test.

H_0 :

H_A :

1.2 [1 point] List ONE assumption required for this test.

1.3 [2 points] Compute the pooled estimate of the overall proportion, $\hat{p}$. Show your work.

Final Answer (round to 2 decimals):

1.4 [1 point] Compute the standard error. Show your work.

Final Answer (round to 2 decimals):

1.5 [1 point] Compute the test statistic. Show your work.

Final Answer (round to 2 decimals):

The research team calculated the p-value below:

```
## [1] 4.273694e-09
```

1.6 [1 point] Interpret the results of the hypothesis test *in the context of the study.* (1 sentence)

Question 2 [6 points]

The UC Berkeley Alternative Meats Lab is reinventing the plant-based burger, designed to be more flavorful and appealing in texture. Their goal is to support environmental health and the prevention of chronic disease through the reduction of red meat consumption.

To test whether their plant-based patty leads to increased consumption, they design a taste-test study with 20 participants. Each participant is served the original version and the new version of the plant-based burger on two separate trays in random order. After each tasting session, the company records the number of grams consumed.

They conduct a paired t-test to evaluate whether the new patty affects how much people eat. The output is shown below:

```
Paired t-test
data: grams_original and grams_new
t = 2.12, df = 19
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
0.3 22.5
sample estimates:
mean of the differences
11.4
```

Q2.1 [1 point] State the null and alternative hypotheses for this t-test using the appropriate notation and context.

Null:

Alternative:

Q2.2 [1 point] Interpret the confidence interval in the context of the question. What does it tell us about the difference in grams consumed?

Q2.2 extra credit [1 point] What would we assume about the p value for this test based on the results presented?

Q2.3 [2 points] Describe an assumption that we discussed in class that must be met for the t-test in this study to be valid and which we could assess with our sample data? Describe how you would conduct this assessment.

Assumption:

Assessment:

Q2.4 [1 point] The data analysis team at the Alternative Meats Lab mistakenly analyzed this data using a two-sample t-test instead of a paired t-test. Explain how this mistake would affect the statistical power of the study and why.

Q2.5 [1 point] Would you expect the standard error in the two-sample analysis to be larger, smaller, or the same as the paired analysis? Explain your answer in the context of the study.